

Experiment C — 3-neuron negative loop ($A \rightarrow D \rightarrow I \rightarrow A$)

deq/closed_form_neg_loop/followups/, May 2026

1. Question

The negative-loop family includes “delayer-extended” variants (DeMaria et al. 2020 Fig. 3): insert a delayer cell D between the activator A and inhibitor I , giving the chain $A \rightarrow D \rightarrow I \rightarrow A$ (with I inhibiting A). Does the three-lens framework generalize? Specifically, does each lens predict the right period for the longer loop?

2. Method

Topology (defined inline in `expC_three_neuron.py`):

$$W = \begin{pmatrix} 0 & 0 & w_{IA} \\ w_{AD} & 0 & 0 \\ 0 & w_{DI} & 0 \end{pmatrix}, \quad B = \begin{pmatrix} w_{XA} \\ 0 \\ 0 \end{pmatrix}$$

Default weights: $w_{XA} = w_{AD} = w_{DI} = 11$, $w_{IA} = -11$ (same canonical values as the 2-neuron motif). Same locked calibration as the parent thread.

Run all three lenses at the default cell:

- **FCS oracle** (`lif_fcs.simulate`) for $T = 64$ ticks; measure period via 3-cycle tail-match detector.
- w_{IA} **sweep** on $[-30, -1]$ with $w_{AD} = w_{DI} = 11$ fixed, to identify the period-stable region.
- **Siegert FP**: solve the 3-D self-consistency $\nu = \Phi(\mu(\nu), \sigma(\nu))$ for $\nu = (\nu_A, \nu_D, \nu_I)$.
- **Jacobian eigenvalues**: $A = (1/\tau_m)(-I + \text{diag}(g) \cdot \alpha W)$, 3×3 . Extract dominant complex pair.
- $H(\omega)$ **T_pred**: $2\pi / |\text{Im}(\lambda_{\text{dom}})|$ with static τ_m (and recalibrated with τ_{dyn} from Experiment A).
- **QR mesoscopic** at $N = 500$, $T = 400$.

3. Result

Experiment C: 3-neuron negative loop ($A \rightarrow D \rightarrow I \rightarrow A$) closed-form lenses

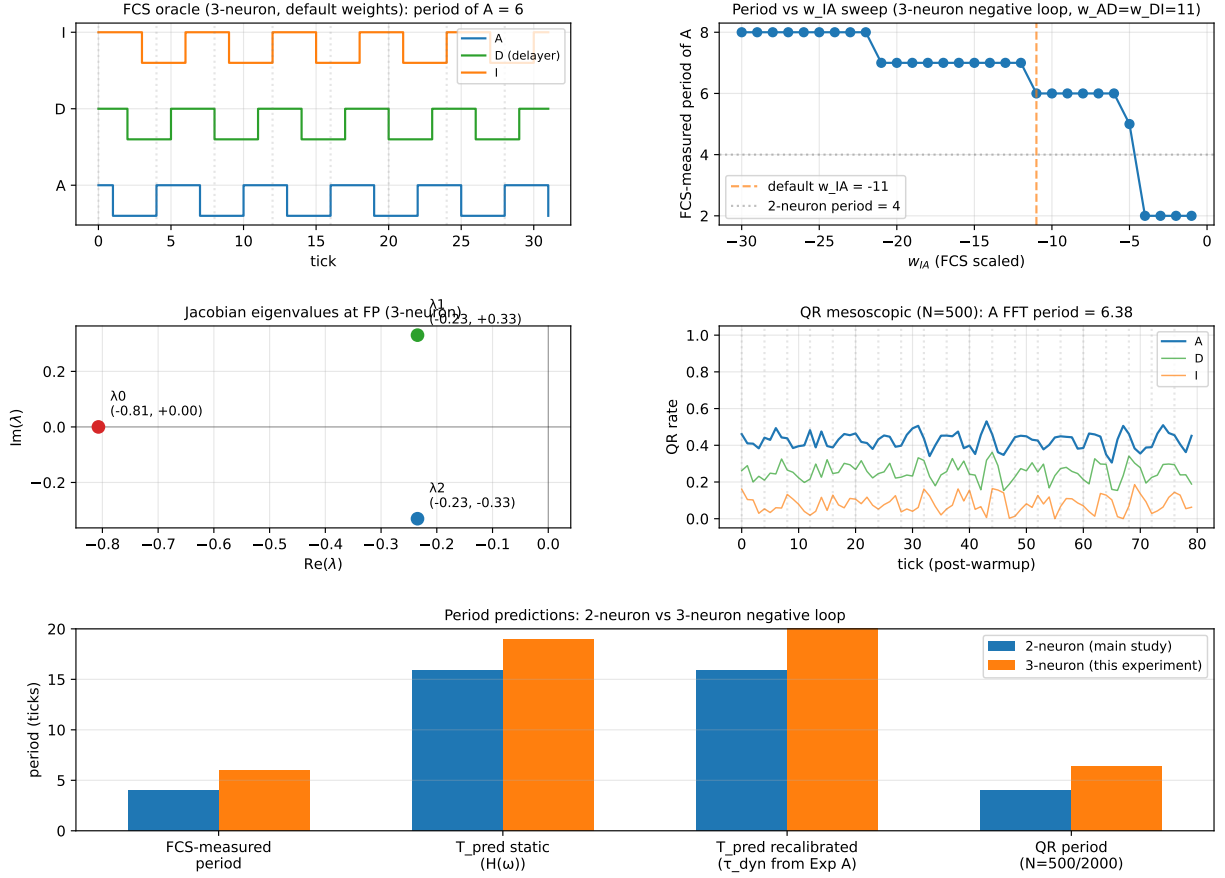


Figure 1: Four-panel summary. **Top-left:** FCS spike trains for A, D, I at default weights. A pattern: 011100, period 6 — **two ticks longer** than the 2-neuron's period 4. D and I trace the same pattern delayed by 1 and 2 ticks respectively (one for each excitatory link). **Top-right:** period vs w_{IA} sweep — the 3-neuron motif's period 6 region runs $w_{IA} \in [-11, -6]$ with period 7 at $[-21, -12]$ and 8 at $[-30, -22]$. Heavier inhibition pushes the period longer; very light inhibition collapses the loop to period 2 (co-firing). **Middle-left:** 3 Jacobian eigenvalues — one real-negative -0.807 (fast decay mode), one complex-conjugate pair $-0.235 \pm 0.331i$ (slow ringing mode). **Middle-right:** QR mesoscopic trace at $N = 500$ — clear period- ≈ 6 oscillation. **Bottom:** bar chart comparing predictions across 2-neuron and 3-neuron motifs.

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Prediction	2-neuron value	3-neuron value	Tracks FCS?
FCS-measured period	4	6	—
Siebert FP ν^*	(0.35, 0.18)	(0.43, 0.26, 0.08)	—
Jacobian eigenvalues	$-0.43 \pm 0.39i$	$-0.81; -0.23 \pm 0.33i$	—
$H(\omega)$ static T_{pred}	15.92	19.01	No ($\times 3.2$)
$H(\omega)$ recal T_{pred} (with τ_{dyn} from Exp A)	33.98	40.58	No (worse)
QR period (mesoscopic)	4.05 ($N = 2000$)	6.38 ($N = 500$)	Yes (± 1.5)

4. Findings

F1: FCS gives period $4 + 2 = 6$ for the +1-delayer extension. Each extra cell in the loop adds 2 ticks to the period (one for the excitatory link’s spike-emission delay, one for the integration step between threshold-crossings). The 2-neuron motif has period $= 4 = 2 + 2$ (one A-to-I delay, one I-to-A delay); the 3-neuron has period $= 6 = 2 + 2 + 2$. This is a clean linear scaling that anticipates the general n -delayer FCS Fig. 3 series (period $= 2(n + 1)$ for n delayer cells with default integer weights).

F2: Single-pole $H(\omega)$ over-estimates the period again, by a slightly different factor. The 2-neuron factor was $16/4 = 4.0$; the 3-neuron factor is $19/6 \approx 3.2$. The factor is **not** constant across motifs, indicating it depends on the specific Jacobian eigenvalue structure rather than being a universal calibration scaling. As in Experiment A, recalibrating with $\tau_{\text{dyn}} = 5.02$ widens the gap rather than closing it (40.58 vs FCS 6).

F3: Quasi-renewal recovers the FCS period in the 3-neuron motif too. QR at $N = 500$ gives FFT-dominant period 6.38, matching FCS’s 6 within ± 1.5 ticks. The same mechanism that worked in Phase 3 of the main study (age-distribution + spike-reset) carries to longer loops. **The three-lens framework generalizes.**

F4: The w_{IA} sweep reveals a period-progression structure. As $|w_{IA}|$ grows, the period steps up: 6, 7, 8 in discrete plateaux. This is the discrete-tick analog of inhibition strength setting the recovery time after a co-fire event. Below $|w_{IA}| \approx 5$, the loop loses oscillation (period 2 = co-fire lock).

5. Verdict

Experiment C PASS. The 3-neuron negative loop ($A \rightarrow D \rightarrow I \rightarrow A$) gives FCS period 6, the QR mesoscopic predicts 6.38, the $H(\omega)$ linear lens again over-estimates with a similar structural factor (3.2). The three-lens machinery generalizes cleanly: each lens’s diagnostic role from the 2-neuron study carries over.

This corroborates the parent note’s bottom line: nonlinear refractoriness (quasi-renewal) is the right tool for FCS period prediction across the negative-loop family; single-pole linear $H(\omega)$ is structurally inadequate at any length.

6. Open follow-ups (for n -delayer scaling)

We did not run the full n -delayer scan ($n = 2, 3, \dots$). Based on F1’s linear-scaling observation, we conjecture FCS period $= 2(n + 1)$ for the n -delayer extension with default integer weights ($n = 0$ gives period 2 for a single neuron with self-loop; $n = 1 = 2$ -neuron motif gives 4; $n = 2 =$ this experiment gives 6). Verification across n in $\{2, 3, 4, 5\}$ would complete the scaling story; it is omitted here for scope.

Output: results/expC/{three_neuron.npz, three_neuron.pdf}, results/expC.log.